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A Closer Look at the Self-Verification Abilities of Large Language Models in Logical Reasoning

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Abstract

Logical reasoning has been an ongoing pursuit in the field of AI. Despite significant advancements made by large language models (LLMs), they still struggle with complex logical reasoning problems. To enhance reasoning performance, one promising direction is scalable oversight, which requires LLMs to identify their own errors and then improve by themselves. Various self-verification methods have been proposed in pursuit of this goal. Nevertheless, whether existing models understand their own errors well is still under investigation. In this paper, we take a closer look at the self-verification abilities of LLMs in the context of logical reasoning, focusing on their ability to identify logical fallacies accurately. We introduce a dataset, FALLACIES, containing 232 types of reasoning fallacies categorized in a hierarchical taxonomy. By conducting exhaustive experiments on FALLACIES, we obtain comprehensive and detailed analyses of a series of models on their verification abilities. Our main findings suggest that existing LLMs could struggle to identify fallacious reasoning steps accurately and may fall short of guaranteeing the validity of self-verification methods. Drawing from these observations, we offer suggestions for future research and practical applications of self-verification methods.[1]

1 Introduction

Logical reasoning is not only a crucial aspect of human intelligence but also one of the long-term pursuits of artificial intelligence (McCarthy, 1989). It is indispensable in intelligent systems, enabling problem-solving, decision-making, and critical thinking. Large language models (LLMs) have recently achieved remarkable advancements in a wide range of tasks (OpenAI, 2023). Being prompted appropriately, LLMs exhibit the emergent ability to "reason" step by step like humans (Wei et al., 2022a,b). Nonetheless, increasing research suggests that LLMs struggle with intricate logical reasoning problems, occasionally producing unfaithful reasoning steps fraught with logical fallacies (Arkoudas, 2023; Liu et al., 2023a; Pan et al., 2023; Hong et al., 2023).

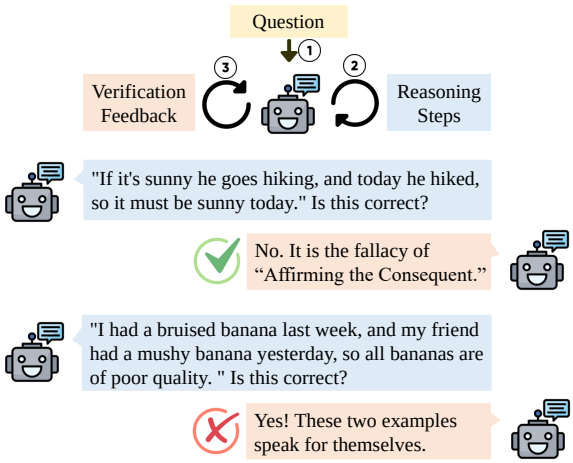

Figure 1: The self-verification approach requires LLMs to identify the fallacious steps in their own reasoning process. However, LLMs might be susceptible to certain types of fallacies and fail to identify them, leading to the potential failure of self-verification.

To tackle this issue, a prevalent and promising approach is scalable oversight (Bowman et al., 2022), where the LLMs could be boosted based on their own evaluation signals (Leike and Sutskever, 2023). In this regard, various strategies of self-verification using LLMs are proposed to enhance reasoning performance (Ling et al., 2023; Weng et al., 2022; Xie et al., 2023; Madaan et al., 2023; Miao et al., 2023). As shown in Figure 1, the LLMs first generate the reasoning process and then self-verify their own output. The verification results are then used to refine the output or further improve the models. Their *assumption* is that LLMs can reliably identify fallacious reasoning steps.

Though empirical evidence demonstrates their

* Work done during the internship at Tencent AI Lab.
[1]Data is available at https://github.com/Raising-hrx/FALLACIES.

preliminary success, a thorough and comprehensive evaluation of the underlying assumption remains unexplored. First, these efforts typically use the performance of the final task (e.g., answer accuracy) to illustrate their effectiveness. However, this is a proxy metric that does not directly reflect the ability of LLMs to identify logical fallacies. LLMs might possibly arrive at the correct answer despite the existence of fallacious intermediate steps (Wei et al., 2022b; Creswell et al., 2023; Lanham et al., 2023). Second, they are usually only concerned with *whether or not* the reasoning step is fallacious, rather than *what type of* fallacy exists in the step, which could be informative for the self-improvement of LLMs. Such an oversimplification precludes a definitive analysis of the LLMs' ability to identify different types of fallacies.

In this paper, we provide a comprehensive evaluation of the verification abilities of LLMs in logical reasoning. Specifically, we collect a dataset, FALLACIES, containing 4,640 reasoning steps for 232 types of fallacies. We evaluate whether LLMs can distinguish between correct and fallacious reasoning steps. Such directed evaluation can provide a more accurate reflection of the verification abilities of LLMs, as it steers away from proxy metrics and delves straight into the actual performance in identifying fallacious steps. Compared to previous datasets, our dataset features more types of fallacies, a larger scale, more fine-grained reasoning, and explicit premises and conclusions. Furthermore, we adopt a hierarchical taxonomy of fallacies, which divides fallacies into two main categories and nine subcategories, allowing for a more systematic approach toward analyzing the verification abilities of LLMs across varying aspects.

We conduct exhaustive experiments on a range of LLMs. First, experimental results show that most LLMs struggle with accurately identifying the fallacious steps. Most LLMs only achieved an overall accuracy rate of less than 80%, suggesting that LLMs could lack sufficient logical verification abilities. Thus, we should be more cautious about the self-verification methods of LLMs. Second, the performance of LLMs can be remarkably imbalanced in different types of fallacies. Most LLMs perform much worse at identifying fallacies related to logical structure than those related to content, pointing toward key directions for improving LLMs' verification and reasoning abilities. Third, we also find that LLMs encounter challenges when it comes to classifying different types of fallacies. Presenting LLMs with the definitions of fallacies does not appear to improve their ability to recognize fallacies. This raises a call for further research to delve into the underlying mechanisms through which LLMs understand reasoning and fallacies. In summary, we present a comprehensive evaluation of the verification abilities of LLMs and highlight their limitations in identifying fallacies, urging the community to apply the self-verification methods with caution.

2 Related Work

2.1 Language Models for Logical Reasoning

Compared with traditional symbol-based logic reasoning systems, using language models to directly reason over natural language is a more flexible and popular approach (Yu et al., 2023; Yang et al., 2023b). Many efforts have been devoted to improving the logical reasoning abilities of language models from different perspectives, including fine-tuning methods (Clark et al., 2020; Dalvi et al., 2021), pre-training methods (Pi et al., 2022; Jiao et al., 2022), and modular methods (Hong et al., 2022; Yang et al., 2022). As the model scale increases, specially designed prompts (e.g., Chain-of-Thought prompts (Wei et al., 2022b)) can elicit the step-by-step reasoning abilities of LLMs, achieving remarkable improvement in multiple reasoning tasks (Chu et al., 2023). However, some studies find that LLMs still struggle with complex logical reasoning problems (Arkoudas, 2023) and could be susceptible to logical fallacies (Payandeh et al., 2023). There is still a lack of comprehensive research to investigate the understanding of LLMs on different logical fallacies.

2.2 Self-Verification with Large Language Models

A prevalent method to enhance the capacity of LLMs is through learning and correction via high-quality verification feedback. This feedback proves instrumental in various aspects of model optimization. For instance, it can be used to fine-tune the models (Ouyang et al., 2022; Scheurer et al., 2023; Chen et al., 2023a; Zelikman et al., 2022; Wang et al., 2023; Bai et al., 2022; Huang et al., 2022). In addition, verification feedback can be utilized to re-rank the model outputs (Lightman et al., 2023; Weng et al., 2022; He et al., 2023; Ni et al., 2023; Ling et al., 2023). It can also be em-

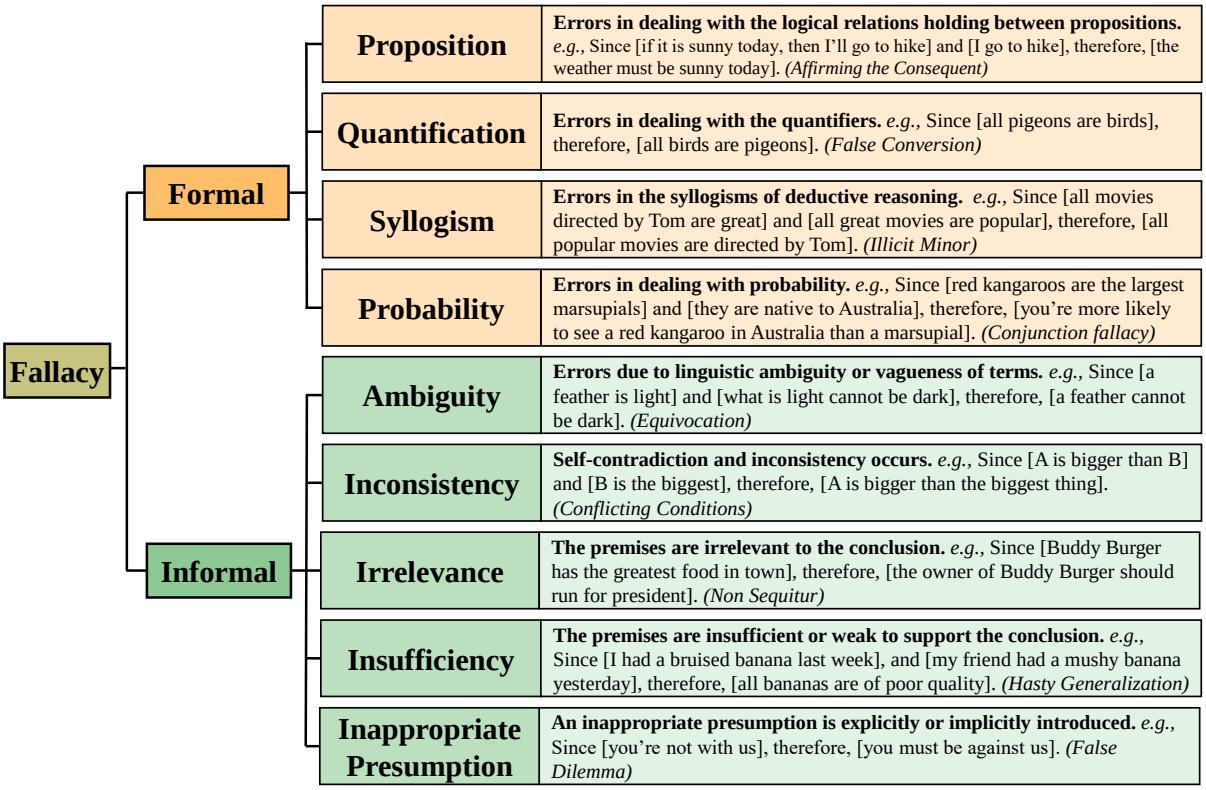

Figure 2: The hierarchical taxonomy of fallacies. For each sub-category, we present its definition and an example of a fallacy within the sub-category. We use square brackets to indicate the premises and conclusions.

ployed to refine the outputs (Madaan et al., 2023; Shinn et al., 2023; Chen et al., 2023b) and guide the generation process of the models (Yao et al., 2023; Xie et al., 2023). A straightforward approach to collecting feedback is to collect directly from humans (Ouyang et al., 2022; Fernandes et al., 2023), but this could prove to be costly and unable to provide instant feedback. Alternatively, collecting feedback from external tools (Gou et al., 2023; Chern et al., 2023) or metrics (Jung et al., 2022) could be more feasible but limited to specific tasks. Thus, some researchers turn to use large language models themselves to provide verification feedback (Ling et al., 2023; Weng et al., 2022; Xie et al., 2023; Madaan et al., 2023; Miao et al., 2023), which is more scalable. Nonetheless, recent papers (Huang et al., 2023; Valmeekam et al., 2023) have raised doubts about the self-verification abilities of the LLMs. For instance, Huang et al. (2023) find that LLMs struggle to self-correct their responses without external feedback. However, they still leave some open questions for subsequent research, such as what exactly the performance of LLMs is to verify a single reasoning step. In this paper, we delve deeply into this subject, critically examining the verification abilities of LLMs from the perspective of logical reasoning.

Category	# Fallacy
Formal	**24**
proposition (prop.)	6
quantification (quant.)	6
syllogism (syl.)	8
probability (prob.)	4
Informal	**208**
ambiguity (amb.)	15
inconsistency (incon.)	3
irrelevance (irrel.)	78
insufficiency (insuf.)	58
inappropriate presumption (inappr.)	54

Table 1: Distribution of 232 fallacies in FALLACIES.

3 FALLACIES

This section outlines the design principles and process of constructing our dataset FALLACIES.

3.1 Design Principles

Covering more comprehensive error types: We propose to evaluate the verification abilities of LLMs on a wider range of types of errors. Relying solely on a single existing dataset of logi-